# How Testing Of Satellites Has Evolved Over The Years

A satellite has to operate in very harsh conditions after launch. Also, building a good satellite requires billions of dollars. Therefore it's crucial to test each satellite properly before it enters space to ensure good performance and safety. This article covers the various types of satellites testing and challenges that may occur during the process

*Radio Check-Out Equipment (RF SCOE) (Credit: https://atos.net)*

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Satellites, one of the long-existing inventions, are exhibiting their muscular and exuberant strength in the right way. For years, satellites have been providing services like navigation, communication, remote sensing, surveillance, earth observations, etc. Since satellites face lots of hazards in space during their operation, efficient technological standards are required to design them.

Today, most of the people depend on satellites in one way or the other, from GPS-based vehicles to weather forecasting to many other things in their lives. Currently, there are almost 1,500 active satellites in orbit around Earth.

Building a good satellite is not cheap. Government or private companies spend millions or even billions of dollars a year on satellites, and it would result in a heavy loss if a satellite fails during or following launch. So, tests and measurements play an important role in the case of satellites.

Tests and measurements are required for components of satellites manufactured for sub-systems and final assembly. Evaluating the performance of satellites is essential even on the ground station terminal. Such smart test and measurement solutions ensure good performance and reliable safety and security on every stage during a satellite launch.

## Inside the world of satellite testing

Extreme vibrations and high acoustic levels exist during satellite launch. Further, a satellite has to operate in very harsh conditions after launch. It has to function in a vacuum while handling high levels of electromagnetic radiation and fluctuation in temperatures.

Companies like NTS maintain a network of facilities across the US to conduct necessary tests to analyse how a satellite responds to a vacuum of space or extreme changes in temperature.

More tests are required for vibration, solar radiation, dealing with the dust of space, or pyroshock, which might occur during the booster separation stage or satellite separation stage from exploding bolts. Sometimes explosive shock might also occur during the booster separation stage. This can even damage circuits,